

Top 4 Hypervisor Companies



VMware by Broadcom

Type 1 hypervisor that holds a roughly 41% market share. Utilized for large enterprise environments.

Pros:

- Hardened security system
 - vSphere Trust Authority, secure boot, encrypted vMotion)
- Large base of tooling and third-party security integrations
- Well-documented CVE history and patching

Cons:

- Broadcom shut down free versions and transitioned to expensive subscriptions
- Licensing volatility is a security/business risk
 - Vendor lock-in
 - Unpredictable pricing shifts

Hyper-V by Microsoft

Type 1 hypervisor that holds a roughly 5.3% market share. Utilized by smaller to mid-size organizations that use the Microsoft ecosystem

Pros:

- Native integration with AD, Defender for Endpoint, and Windows security tooling
- Easier compliance mapping for Microsoft-centric businesses
- Frequent patching via normal Windows Update channels

Cons:

- No standalone Hyper-V Server, must run inside Windows Server
- Management interface is clunkier than other hypervisors and payment is required for extra tooling
- Need to pay for Windows Server licensing

Proxmox Server Solutions

- Proxmox VE is an open-source, Type-1 hypervisor based on KVM (Kernel-based Virtual Machine)
- Has become massively popular for building large-scale "cyber ranges" and home hacking labs

Pros:

- Completely free and open-source; allows you to run full VMs and lightweight Linux Containers (LXC) on the same host;
- Manage VMs, storage, and networking from a clean, web-based graphical interface.
- Includes a built-in firewall that allows you to define security rules for all hosts in a cluster

Cons:

- Updating the system between major software versions can be messy and lacks automation.

KVM (Kernel-based Virtual Machine)

- KVM is the raw virtualization module baked directly into the Linux kernel. It turns any standard Linux machine into a Type-1 hypervisor

Pros:

- Because it is built directly into the Linux kernel, it runs incredibly fast with almost no overhead.
- From a threat-modeling perspective, it is highly secure. VMs run as unprivileged, user-mode processes.

Cons:

- raw KVM is heavily reliant on the Linux command-line interface (CLI). It requires significant technical expertise to configure networking and storage manually.
- Basic operational tasks, such as resizing the disk dimensions of an existing VM are often not user-friendly